

Effect of the kappa correction in RBLapSum

Research note: `through_rank_kappa` versus plain `through_rank` at $b_0 = 0$

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1 Purpose and summary of conclusions

RBLapSum’s boundary surrogate can be applied in two forms. In plain `through_rank`, each candidate’s score gradient is the kernel-weighted boundary term itself. In `through_rank_kappa`, the same term is additionally passed through the LapSum rank-one Jacobian at the boundary,

$$g = a - q \sum_i a_i, \quad q_i = \kappa_i / \sum_j \kappa_j,$$

which removes the common score-translation mode. This note tests whether the kappa correction improves convergence, using matched training runs at a boundary floor of $b_0 = 0$ (the new default; the earlier campaigns used $b_0 = 0.1$).

Conclusions:

- **The kappa correction is decisively better in every tested configuration.** Plain `through_rank` diverges within the first ~ 1700 steps in all four configurations with $K \geq 64$, and in the two $K = 32$ configurations where it survives it converges 0.085–0.164 nats worse than `through_rank_kappa` while showing repeated large excursions along the way.
- **With the correction, five of six configurations converge**, at final validation CE 1.46–1.55; the exception is $(K, J) = (64, 64)$ at $T = 1$, which diverged at step 8680. The $K = 64$, $T = 1$ row is the known marginal row of this surrogate.
- **The boundary floor does not matter in these regimes.** The four `through_rank_kappa` configurations that were previously trained at $b_0 = 0.1$ reproduce at $b_0 = 0$ within ± 0.005 nats, in line with the earlier measurement that the mean boundary stays 8–40 $\times$ above the old floor.

2 Setup

Twelve runs, all under full magnitude–direction decoupling with the same architecture, data, optimizer, schedules, and seed as the concurrent MD-versus-MD-init campaign (8×768 , `abs_topk` over $N = 1536$, `residual_out`, batch 96×512 , 20k steps). The grid pairs the two gradient modes at

$$(K, J, T) \in \{(32, 32, 1), (32, 480, 1), (64, 64, 1), (64, 448, 1), (128, 128, 2), (128, 384, 2)\},$$

with $T = 2$ for the $K = 128$ rows because $K = 128$, $T = 1$ is known to diverge under every regime. All runs use $b_0 = 0$; the run names carry a `b00` marker. Divergence is detected and the run stopped by the same guard as in the companion note (non-finite CE, or CE far above its own best for a sustained window).

3 Results

(K, J)	T	through_rank_kappa	through_rank
(32, 32)	1	1.5533	1.6382
(32, 480)	1	1.4918	1.6558
(64, 64)	1	diverged (8680)	diverged (1580)
(64, 448)	1	1.4776	diverged (1620)
(128, 128)	2	1.4792	diverged (1580)
(128, 384)	2	1.4615	diverged (1680)

Final validation CE after 20k steps, or the step at which the run was stopped as diverged.

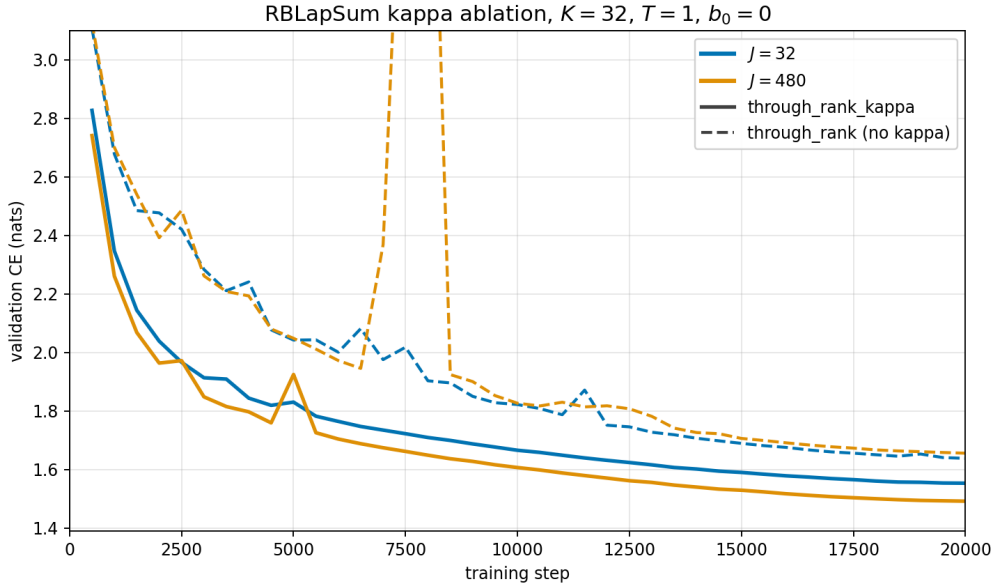


Figure 1: $K = 32$, $T = 1$: both modes survive. The plain `through_rank` runs (dashed) sit ~ 0.1 nats above their kappa counterparts throughout training and undergo repeated large excursions, including a recovered burst of the $J = 480$ run around steps 7000–8500.

Comparison with the earlier $b_0 = 0.1$ runs. Four of the kappa configurations have direct twins in the MD-versus-MD-init campaign, which ran at $b_0 = 0.1$ but is otherwise identical:

(K, J)	T	$b_0 = 0.1$	$b_0 = 0$	difference
(32, 32)	1	1.5482	1.5533	+0.0051
(32, 480)	1	1.4923	1.4918	−0.0005
(128, 128)	2	1.4778	1.4792	+0.0014
(128, 384)	2	1.4602	1.4615	+0.0013

The differences are within run-to-run noise, confirming that the earlier results were not shaped by the floor.

One historical contrast is worth recording: at $(K, J) = (64, 448)$, $T = 1$, the original $b_0 = 0.1$ campaign run destabilized near step 3940, while the present $b_0 = 0$ run converged cleanly. Given the four replication pairs above, this should be read as the known run-to-run stochasticity of divergence in a marginal row rather than as an effect of the floor; the same row also splits here between its $J = 448$ survivor and its $J = 64$ death.

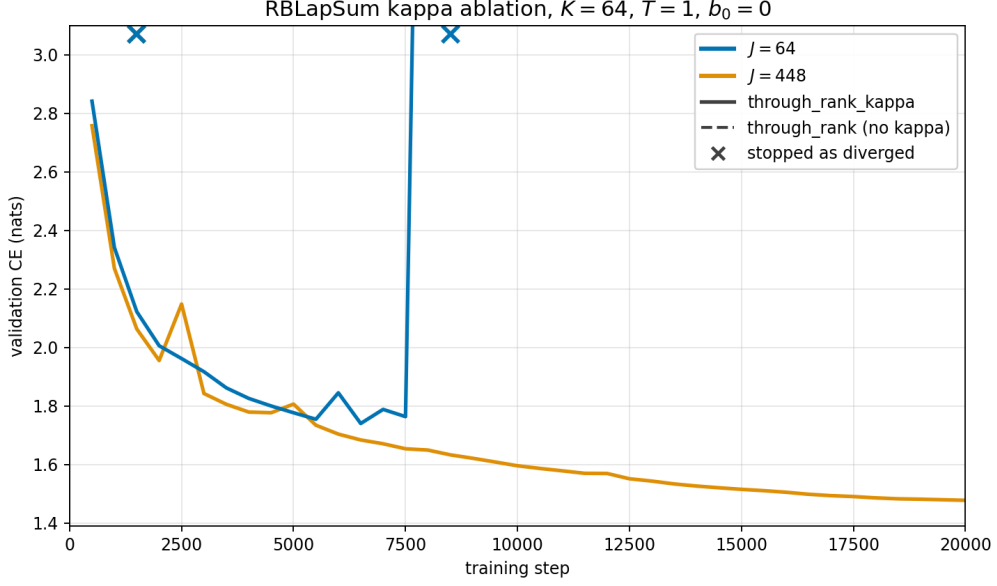


Figure 2: $K = 64$, $T = 1$: both `through_rank` runs diverge within the first ~ 1600 steps (crosses at the top edge). With the kappa correction, $J = 448$ converges to 1.4776 while $J = 64$ survives to step 8680 before a terminal burst.

4 Conclusion

The kappa correction is not an optional refinement of the `through_rank` surrogate; in these setups it is what makes the surrogate usable. Without it, training survives only in the most conservative row tested ($K = 32$, $T = 1$), and even there at a 0.085–0.164 nat penalty with visibly bursty dynamics. With it, every configuration except the marginal $(64, 64)$, $T = 1$ cell converges to 1.46–1.55. All cells are single runs at one seed, so individual divergences carry the usual stochastic caveat; the mode-level separation, however, is uniform across all six configurations.

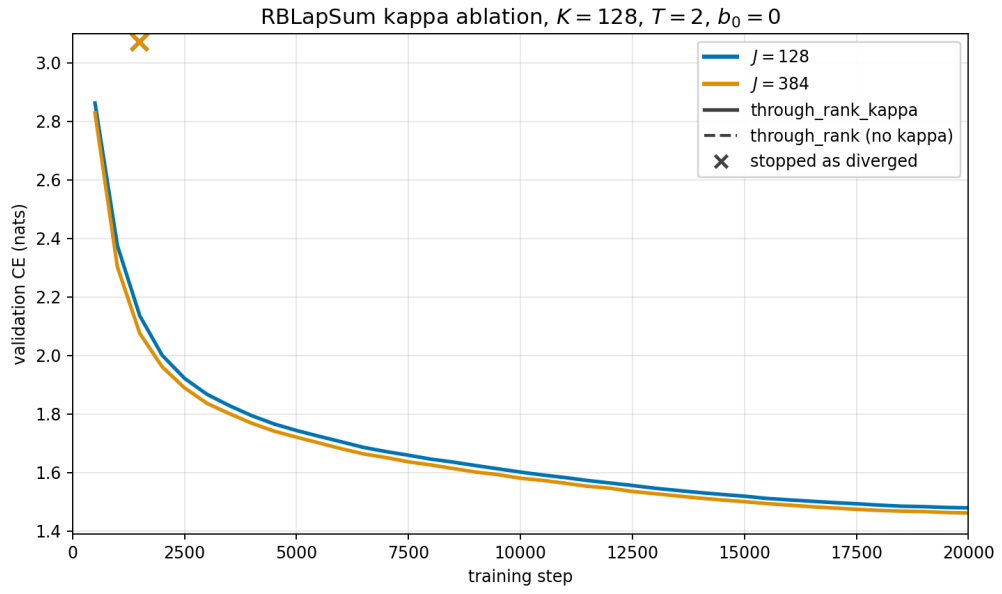


Figure 3: $K = 128, T = 2$: both `through_rank` runs diverge within the first ~ 1700 steps; both kappa runs converge.